In [1]:
```python
import pandas as pd
import numpy as np
import warnings
warnings.filterwarnings("ignore")
import matplotlib.pyplot as plt
import seaborn as sns
```

In [2]:
```python
df = pd.read_csv("mymoviedb.csv",lineterminator='\n')
```

In [3]:
```python
# df["Popularity"].plot(kind = "kde")
```

In [4]:
```python
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9827 entries, 0 to 9826
Data columns (total 9 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Release_Date       9827 non-null   object
 1   Title              9827 non-null   object
 2   Overview           9827 non-null   object
 3   Popularity         9827 non-null   float64
 4   Vote_Count         9827 non-null   int64
 5   Vote_Average       9827 non-null   float64
 6   Original_Language  9827 non-null   object
 7   Genre              9827 non-null   object
 8   Poster_Url         9827 non-null   object
dtypes: float64(2), int64(1), object(6)
memory usage: 691.1+ KB
```

Data_Wrangling

1. data Gathering.
2. identifying problems in a data.
3. solving the problem which we have discovered in step no. 2

In [5]:
```python
df.duplicated().sum()
```

Out[5]:
```
0
```

In [6]:
```python
df.isnull().sum()
```

Out[6]:
```
Release_Date         0
Title                0
Overview             0
Popularity           0
Vote_Count           0
Vote_Average         0
Original_Language    0
Genre                0
Poster_Url           0
dtype: int64
```

In [7]:
```python
df["Genre"]
```

Out[7]:
```
0       Action, Adventure, Science Fiction
1                 Crime, Mystery, Thriller
2                                 Thriller
3       Animation, Comedy, Family, Fantasy
4         Action, Adventure, Thriller, War
                       ...
9822                          Drama, Crime
9823                                Horror
9824              Mystery, Thriller, Horror
9825                 Music, Drama, History
9826           War, Drama, Science Fiction
Name: Genre, Length: 9827, dtype: object
```

In [8]:
```python
df.describe()
```

Out[8]:

	Popularity	Vote_Count	Vote_Average
count	9827.000000	9827.000000	9827.000000
mean	40.326088	1392.805536	6.439534
std	108.873998	2611.206907	1.129759
min	13.354000	0.000000	0.000000
25%	16.128500	146.000000	5.900000
50%	21.199000	444.000000	6.500000
75%	35.191500	1376.000000	7.100000
max	5083.954000	31077.000000	10.000000

identifying problems in a data.(Data Preprocessing) 1) data type of Realese_data column need to change from object to datetime. 2) there are 3 columns that we need to drop bez they are of no use to us. ==> Overview, Original_Language, Poster_Url .(As per 5 questions that we are going to solve.) 3) Genre have some white spaces after the comma(,) that we need to trim out. 4) there are noticable Outliers in Popularity Column# Good points 1) Good this is there are no duplicate rows avaliable in our data set. 2) Also no null values are present in dataset.

In [9]:
```python
df["Vote_Average"].describe()
```

Out[9]:
```
count    9827.000000
mean        6.439534
std         1.129759
min         0.000000
25%         5.900000
50%         6.500000
75%         7.100000
max        10.000000
Name: Vote_Average, dtype: float64
```

Cleaning the Problems that we have identified in step 2

In [10]:
```python
# changing data type of Release_data


df["Release_Date"] = pd.to_datetime(df["Release_Date"])
```

In [11]:
```python
# Dropping 3 columns Overview, Original_Language, Poster_Url


df.drop(columns = ["Overview", "Original_Language","Poster_Url"],axis = 1, inplace = True)
```

In [12]:
```python
""" Removing the white spaces which are avaliable in Genre and we need to split the genre in a catgorical column.
each category will have different line and it will increase the number of lines in dataset.
"""

df["New_Genre"] = df["Genre"].str.split(", ")
df = df.explode("New_Genre").reset_index(drop = True)
```

We have successfully removed the white space problem. and dataset shape has increased to ==> 25793 rows

end of Data_wrangling

In [13]:
```python
# Now extracting year form Realese_Date column, it will add a new column in our dataset
df["Year"] = df["Release_Date"].dt.year
```

In [14]:
```python
df.head()
```

Out[14]:

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre	New_Genre	Year
0	2021-12-15	Spider-Man: No Way Home	5083.954	8940	8.3	Action, Adventure, Science Fiction	Action	2021
1	2021-12-15	Spider-Man: No Way Home	5083.954	8940	8.3	Action, Adventure, Science Fiction	Adventure	2021
2	2021-12-15	Spider-Man: No Way Home	5083.954	8940	8.3	Action, Adventure, Science Fiction	Science Fiction	2021
3	2022-03-01	The Batman	3827.658	1151	8.1	Crime, Mystery, Thriller	Crime	2022
4	2022-03-01	The Batman	3827.658	1151	8.1	Crime, Mystery, Thriller	Mystery	2022

NOW APPLYING LABELS ON VOTE_AVERAGE COLUMN 2) give labels to vote_average columns i.e ==> 0.0 - 5.9: "Below Average" 5.9 - 6.5: "Average" 6.5 - 7.1: "Above Average" 7.1 - 10.0: "Popular" or "High"

In [15]:
```python
def label(val):
    if val >=0 and val <5.9:
        return "Below Average"
    elif  val >=5.9 and val < 6.5:
        return "Average"
    elif  val >=6.5 and val < 7.1:
        return "Above Average"
    elif  val >=7.1 and val <= 10:
        return "Popular"
    else :
        return val



df["Rating"] = df["Vote_Average"].apply(label)
df["Rating"].value_counts()
```

Out[15]:
```
Rating
Popular          7560
Above Average    6714
Average          6134
Below Average    5385
Name: count, dtype: int64
```

In [16]:
```python
# casting Rating column from object to category

df["Rating"] = df["Rating"].astype("category")
```

In [17]:
```python
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25793 entries, 0 to 25792
Data columns (total 9 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   Release_Date  25793 non-null  datetime64[ns]
 1   Title         25793 non-null  object
 2   Popularity    25793 non-null  float64
 3   Vote_Count    25793 non-null  int64
 4   Vote_Average  25793 non-null  float64
 5   Genre         25793 non-null  object
 6   New_Genre     25793 non-null  object
 7   Year          25793 non-null  int32
 8   Rating        25793 non-null  category
dtypes: category(1), datetime64[ns](1), float64(2), int32(1), int64(1), object(3)
memory usage: 1.5+ MB
```

Data Visualization

1) what is the most frequent genre of movies is released on NETFLIX PLATFORM?

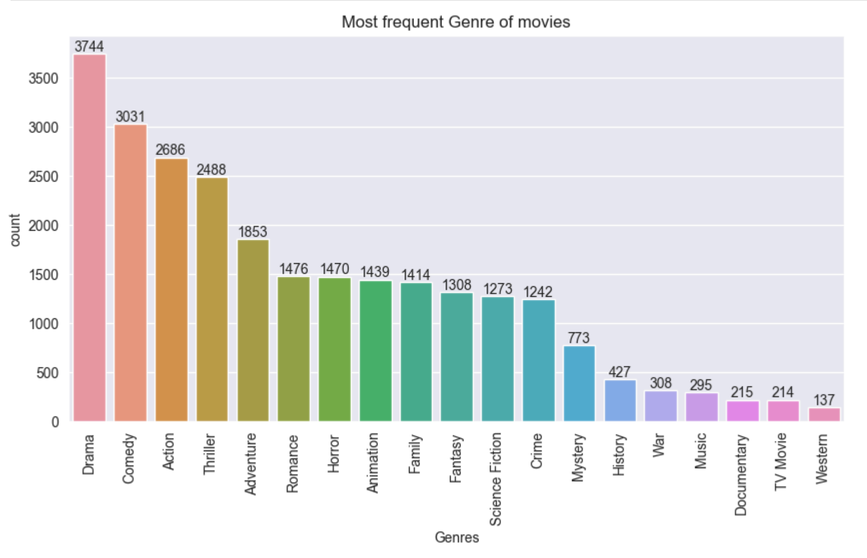
In [18]:
```python
data_genre = df["New_Genre"].value_counts().sort_values(ascending = False).reset_index()
data_genre
```

Out[18]:

	New_Genre	count
0	Drama	3744
1	Comedy	3031
2	Action	2686
3	Thriller	2488
4	Adventure	1853
5	Romance	1476
6	Horror	1470
7	Animation	1439
8	Family	1414
9	Fantasy	1308
10	Science Fiction	1273
11	Crime	1242
12	Mystery	773
13	History	427
14	War	308
15	Music	295
16	Documentary	215
17	TV Movie	214
18	Western	137

In [19]:
```python
sns.set_style("darkgrid")
```

In [20]:
```python
plt.figure(figsize = (10,5))
ax = sns.barplot(data = data_genre, x = "New_Genre", y = "count" )
ax.bar_label(ax.containers[0])
plt.xlabel("Genres")
plt.title("Most frequent Genre of movies")
plt.xticks(rotation = 90)
plt.show()
```



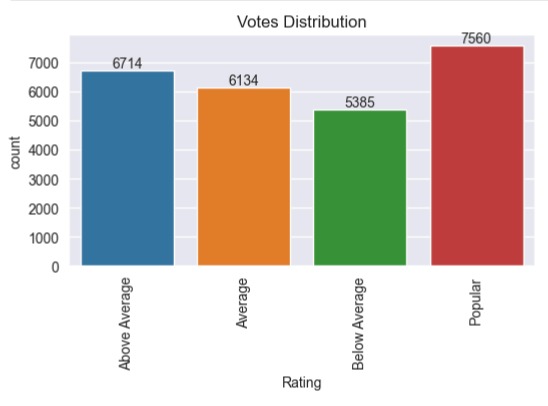
Q. 2) which genre has the highest votes in Rating column?

In [21]:
```python
df.head(2)
```

Out[21]:

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre	New_Genre	Year	Rating
0	2021-12-15	Spider-Man: No Way Home	5083.954	8940	8.3	Action, Adventure, Science Fiction	Action	2021	Popular
1	2021-12-15	Spider-Man: No Way Home	5083.954	8940	8.3	Action, Adventure, Science Fiction	Adventure	2021	Popular

In [22]:
```python
data_rating = df["Rating"].value_counts().sort_values(ascending = False).reset_index()
data_rating
plt.figure(figsize = (6,3))
ax = sns.barplot(data = data_rating, x = "Rating", y = "count" )
ax.bar_label(ax.containers[0])
plt.xlabel("Rating")
plt.title("Votes Distribution")
plt.xticks(rotation = 90)
plt.show()
```



In [23]:
```python
q_2 = df.groupby(["Rating", "New_Genre"]).agg({"Title":"count"}).sort_values(by = "Title", ascending =False).reset_index().head(1)
```

Q. 3 which movies got the highest popularity? what's it's genre

In [24]:
```python
df[df["Popularity"] == df["Popularity"].max()]
```

Out[24]:

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre	New_Genre	Year	Rating
0	2021-12-15	Spider-Man: No Way Home	5083.954	8940	8.3	Action, Adventure, Science Fiction	Action	2021	Popular
1	2021-12-15	Spider-Man: No Way Home	5083.954	8940	8.3	Action, Adventure, Science Fiction	Adventure	2021	Popular
2	2021-12-15	Spider-Man: No Way Home	5083.954	8940	8.3	Action, Adventure, Science Fiction	Science Fiction	2021	Popular

Q. 4 which movies got the lowest popularity? what's it's genre

In [25]:
```python
df[df["Popularity"] == df["Popularity"].min()]
```

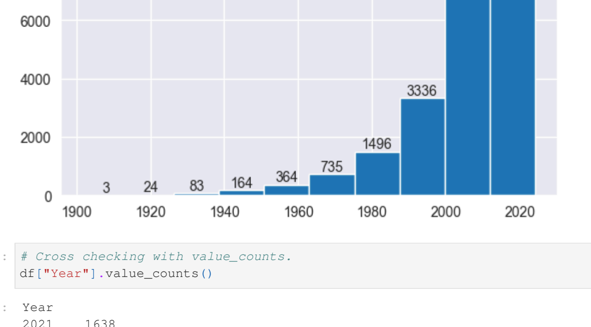
Out[25]:

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre	New_Genre	Year	Rating
25787	2021-03-31	The United States vs. Billie Holiday	13.354	152	6.7	Music, Drama, History	Music	2021	Above Average
25788	2021-03-31	The United States vs. Billie Holiday	13.354	152	6.7	Music, Drama, History	Drama	2021	Above Average
25789	2021-03-31	The United States vs. Billie Holiday	13.354	152	6.7	Music, Drama, History	History	2021	Above Average
25790	1984-09-23	Threads	13.354	186	7.8	War, Drama, Science Fiction	War	1984	Popular
25791	1984-09-23	Threads	13.354	186	7.8	War, Drama, Science Fiction	Drama	1984	Popular
25792	1984-09-23	Threads	13.354	186	7.8	War, Drama, Science Fiction	Science Fiction	1984	Popular

Q. 5 Which Year has the most filmed movies?

In [26]:
```python
ax = df["Year"].hist()
ax.bar_label(ax.containers[0])
plt.title("Most movies filled year")
```

Out[26]:
```
Text(0.5, 1.0, 'Most movies filled year')
```



In [27]:
```python
# Cross checking with value_counts.
df["Year"].value_counts()
```

Out[27]:
```
Year
2021    1638
2018    1387
2017    1365
2019    1272
2016    1212
        ...
2024       4
1902       3
1925       2
1929       2
1930       2
Name: count, Length: 102, dtype: int64
```

Conclusion

Conclusion : - Q. 1) What is the most frequent genre of movies released on NETFLIX PLATFORM? DRAMA is the most frequent genre that is appearing in our dataset. it has appeared more than 14% than the other genres. Q. 2) Which genre has the highest votes in the Rating column? As per calculations, the popular category is at the top. and again drama is appearing at the top and it is more popular among the fans. Q3: What movie got the highest popularity ? What's its genre ? Spider-Man: No Way Home has the highest popularity rate in our dataset and it has genres of Action , Adventure and Science Fiction . Q4: What movie got the lowest popularity ? What's its genre ? The United States, 'thread' has the highest lowest rate in our dataset and it has genres of music , drama , 'war', 'sci-fi' and history. Q5: Which year has the most filmed movies? The year 2021 has the highest filming rate in our dataset.

END OF THE PROJECT, THANK YOU

In [ ]:

In [ ]: